

Files

Introduction

As users of computer systems, we enter data as we are instructed to do, such as typing in our identification code at a cash machine or perhaps filling out a form with our name and address. But data cannot be dumped helter-skelter into a computer. Some computer professional - probably a programmer or systems analyst - has to plan how data from users will be received, organised, and stored, and also in what manner data will be processed by the computer.

What is a File?

A *file* is an organised collection of data. The term file may be used in reference to a database - where a file is equivalent to a table - although any data can be stored in a file.

Logical and Physical Files

A Physical File refers to the way the file is stored on the hardware (e.g. on a disk it would be stored in blocks, which are made up of tracks and sectors). A Logical File is how the programmer structures the file (e.g. in a series of records). A return character can be used to illustrate the difference. To the hardware it is simply a binary code that needs to be stored. To the programmer it might mean “move down a line and back to the beginning of the line” in a text file.

Transaction and Master Files

When data processing takes place generally two types of files are used; transaction files and master files.

- Transaction files keep a record of all transactions that have taken place - usually over a set period. For example a bank machine may keep a record of all transactions in the last 24 hours.
- The master file, however, contains the more permanent data, which is kept up to date by the transactions. So for the above example the master files will contain data about the customers and their balances, and the transaction files will contain information that enables the balance to be kept up-to-date.

Backup and Recovery

How often a file is used, how large the file is, and how important the data is determines the method and frequency of backup. A backup system needs to be in place such that, in the event of failure, the recovery system will mean no or minimal loss of data.